

Supply Chain Control Tower

End-to-End Integration, Dimensional Modeling, and Business Intelligence Deliverables

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CHAPTER 1: INTRODUCTION

1.1 Background

Modern supply chains have evolved into highly interconnected networks involving suppliers, manufacturers, distributors, logistics providers, warehouses, retailers, and end customers operating across multiple geographical regions. Globalization, e-commerce growth, increasing customer expectations, and geopolitical uncertainty have significantly increased the complexity of supply chain operations.

Organizations are no longer competing solely on product quality or price; they are increasingly competing on their ability to deliver products efficiently, reliably, and at the right cost. The rapid growth of digital commerce has fundamentally changed customer expectations regarding product availability and delivery timelines. Customers now expect real-time order visibility, shorter delivery windows, flexible shipping options, and seamless post-purchase experiences. Failure to meet these expectations can result in customer dissatisfaction, revenue loss, and damage to brand reputation.

At the same time, supply chains are exposed to numerous risks including transportation delays, demand fluctuations, inventory shortages, supplier disruptions, port congestion, and volatile fuel prices. The impact of these disruptions often propagates across multiple stages of the supply chain, creating significant operational and financial consequences for organizations. Traditional reporting systems are often unable to provide the

visibility and responsiveness required to manage these challenges effectively. Many organizations continue to rely on siloed operational systems that generate fragmented information, making it difficult for decision-makers to obtain a comprehensive view of supply chain performance.

1.2 Supply Chain Challenges

Modern supply chains face several critical challenges that directly influence operational efficiency and profitability.

- **Lack of End-to-End Visibility:** Supply chain data is often distributed across multiple systems including Enterprise Resource Planning (ERP) platforms, Warehouse Management Systems (WMS), Transportation Management Systems (TMS), and Customer Relationship Management (CRM) applications. This fragmentation prevents organizations from obtaining a unified view of operations.
- **Delivery Delays and Service Failures:** Transportation disruptions, inefficient routing decisions, and inaccurate demand planning frequently result in delayed deliveries. Such delays negatively impact customer satisfaction and increase operational costs.
- **Inventory Imbalances:** Organizations frequently experience both overstocking and stockouts. Excess inventory increases holding costs while insufficient inventory leads to lost sales opportunities and customer dissatisfaction.
- **Demand Volatility:** Customer demand patterns have become increasingly unpredictable due to seasonality, promotions, economic changes, and external events. Traditional forecasting approaches often struggle to respond effectively to these fluctuations.
- **Profitability Pressure:** Competitive markets often encourage aggressive discounting strategies. While discounts may stimulate demand, excessive discounting can significantly reduce profit margins and negatively affect long-term business sustainability.
- **Decision-Making Delays:** Without centralized analytical systems, organizations often rely on historical reports rather than real-time operational intelligence, reducing their ability to respond proactively to emerging issues.

1.3 Supply Chain Control Towers

A Supply Chain Control Tower is an integrated analytical platform designed to provide centralized visibility, monitoring, and decision support across the entire supply chain network. Control towers consolidate operational data from multiple sources and transform it into actionable business intelligence through dashboards, alerts, and predictive analytics.

The primary objective of a control tower is not merely to report historical performance but to enable proactive management of supply chain operations. By providing a unified view of logistics performance, customer demand, inventory movement, and financial outcomes, organizations can identify issues earlier and implement corrective actions more effectively.

Modern supply chain control towers typically focus on five key areas:

1. Executive performance monitoring.
2. Logistics and delivery performance.
3. Customer and market intelligence.
4. Product and inventory management.
5. Financial and profitability analysis.

The increasing adoption of data-driven decision-making has transformed control towers from optional reporting tools into critical components of modern supply chain management strategies.

1.4 Problem Statement

Despite the availability of large volumes of operational data, many organizations continue to face difficulties in converting raw transactional information into actionable insights. Data frequently exists in isolated systems and is often presented in formats unsuitable for executive decision-making. The absence of integrated analytical capabilities limits the organization's ability to identify operational bottlenecks, monitor delivery performance, understand customer behavior, optimize inventory allocation, and evaluate profitability drivers.

This project addresses these challenges by developing an end-to-end Supply Chain Control Tower capable of integrating operational data into a centralized analytical framework. The proposed solution focuses on improving visibility across sales performance, logistics operations, customer behavior, product demand, inventory movement, and financial performance.

1.5 Project Objectives

The primary objectives of this project are:

- To design and implement an end-to-end analytical pipeline for supply chain decision support.
- To transform raw transactional data into business-ready analytical datasets through preprocessing and feature engineering.
- To develop a dimensional data warehouse capable of supporting complex analytical queries.
- To create executive dashboards that provide visibility into operational, customer, inventory, and financial performance.
- To identify key operational risks and business opportunities using data-driven insights.
- To provide actionable recommendations that improve supply chain efficiency and profitability.

1.6 Expected Business Impact

The implementation of a Supply Chain Control Tower provides organizations with significant operational and strategic benefits. Improved visibility enables managers to identify bottlenecks earlier and respond proactively to emerging issues. Enhanced delivery monitoring improves customer service performance and reduces

transportation-related disruptions. Customer segmentation and market analysis support more effective commercial strategies, while inventory and demand analytics improve resource allocation decisions.

Financial analytics enable organizations to better understand the relationship between pricing, discounts, and profitability, supporting more sustainable growth strategies. Ultimately, the proposed solution transforms fragmented operational data into a unified decision-support system capable of improving supply chain resilience, operational efficiency, and business performance.

1.7 Chapter Summary

This chapter introduced the growing complexity of modern supply chains and highlighted the challenges organizations face in achieving visibility and operational efficiency. The concept of Supply Chain Control Towers was discussed as a strategic response to these challenges. The problem statement, project objectives, and expected business impact were subsequently defined, establishing the foundation for the remaining chapters of the report.

CHAPTER 2: DATASET DESCRIPTION AND UNDERSTANDING

2.1 Dataset Source

The dataset utilized in this project is the DataCo Supply Chain Dataset, a publicly available transactional dataset designed for supply chain analytics and business intelligence applications. The dataset contains detailed information related to customer orders, product sales, logistics operations, shipping performance, and financial outcomes across multiple geographical markets.

Unlike traditional academic datasets that focus exclusively on either sales or logistics, the DataCo dataset integrates information from multiple business functions, making it particularly suitable for end-to-end supply chain analysis. The presence of customer, product, geographic, operational, and financial attributes allows the dataset to support cross-functional analysis and executive decision-making. The dataset provides a realistic representation of supply chain operations encountered in retail and distribution organizations.

The dataset captures operations across multiple geographical regions including:

- Africa
- Europe
- LATAM
- Pacific Asia
- USCA

The transactional granularity of the dataset enables detailed analysis at the order level while still supporting aggregation for executive reporting.

2.3 Dataset Structure and Variable Categories

The available variables were classified into six primary business categories to simplify downstream analytics and dimensional modeling.

Customer Attributes

Customer-related variables provide information about customer identity, segmentation, and geographical distribution. Examples include: `customer_id`, `customer_segment`, `customer_city`, `customer_state`, `customer_country`, and `customer_zipcode`. These variables support customer segmentation analysis and market intelligence.

Product Attributes

Product variables describe the products purchased by customers and their associated classifications. Examples include: `product_card_id`, `product_name`, `product_price`, `category_id`, and `department_name`. These variables support product analytics and demand analysis.

Order Attributes

Order variables capture the transactional details of customer purchases. Examples include: `order_id`, `order_item_quantity`, `order_item_discount`, `order_item_discount_rate`, `sales`, and `order_item_total`. These attributes form the basis for revenue and demand analysis.

Logistics Attributes

Logistics variables describe shipping operations and delivery outcomes. Examples include: `shipping_mode`, `days_for_shipping_real`, `days_for_shipment_scheduled`, `delivery_status`, and `late_delivery_risk`. These attributes support operational monitoring and logistics performance measurement.

Financial Attributes

Financial variables capture profitability and commercial performance. Examples include: `benefit_per_order`, `sales_per_customer`, `order_profit_per_order`, and `order_item_profit_ratio`. These variables enable profitability analysis and pricing strategy evaluation.

Geographic Attributes

Geographic variables support regional and market-level analysis. Examples include: `market`, `order_region`, `order_country`, `order_city`, `latitude`, and `longitude`. These attributes support geographic visualization and market analysis.

2.4 Data Quality Assessment

Before analytical processing, the dataset was subjected to extensive quality assessment procedures.

- **Missing Value Analysis:** The dataset was examined for null values across all attributes. Missing values were evaluated individually to determine whether imputation, deletion, or retention represented the most appropriate treatment strategy.
- **Duplicate Analysis:** Duplicate records can distort aggregate metrics and lead to incorrect business conclusions. Duplicate detection procedures were performed to ensure each transaction represented a unique business event.
- **Data Type Validation:** Several attributes required conversion into appropriate formats, such as conversion of date strings into datetime objects, and conversion of numerical attributes into integer and floating-point formats, as well as validation of categorical variables.

Encoding Challenges

During preprocessing, encoding issues were encountered due to the presence of non-ASCII characters in product and customer information. Examples included accented European characters and special symbols in product descriptions. These issues were resolved during preprocessing to ensure successful loading into the SQL database.

2.5 Dataset Limitations

Although comprehensive, the dataset does not contain all information required for a complete digital supply chain twin. The primary limitations include:

- **Absence of Inventory Data:** The dataset does not provide current inventory levels, warehouse stock positions, safety stock values, or reorder points. As a result, inventory optimization was approximated through demand analysis rather than actual stock movement analysis.
- **Absence of Supplier Information:** Information regarding suppliers and procurement timelines is omitted.
- **Partial 2018 Data:** The dataset contains records for 2018 only until January. Including this incomplete year in annual trend analysis would have introduced misleading conclusions. Consequently, 2018 was excluded from year-over-year trend comparisons while remaining available for operational analysis.

2.6 Suitability for Supply Chain Analytics

Despite the identified limitations, the dataset remains highly suitable for the development of a Supply Chain Control Tower due to several factors. First, it integrates operational and financial perspectives within a single dataset. Second, it captures multiple stages of the order fulfillment process including customer ordering behavior, shipping performance, and financial outcomes. Third, its global geographical coverage enables market-level analysis and executive reporting. Finally, the dataset supports both operational analytics and strategic decision-making, making it an ideal candidate for business intelligence applications.

CHAPTER 3: DATA ENGINEERING AND FEATURE ENGINEERING

3.1 Overview of the Data Engineering Pipeline

The objective of the data engineering stage was to transform raw transactional data into a structured and analysis-ready format capable of supporting business intelligence applications and dimensional warehousing. The original dataset consisted of raw order transactions containing customer, product, logistics, and financial information. While suitable for storage, the raw data lacked several business-oriented attributes required for meaningful analytical reporting.

The data engineering pipeline consisted of four primary stages:

1. Data extraction and ingestion.
2. Data cleaning and preprocessing.
3. Feature engineering and business rule generation.
4. Export to the analytical data warehouse.

The transformation process can be represented linearly as: **Raw CSV Dataset** → **Python Data Cleaning** → **Feature Engineering** → **Analytical Dataset** → **MySQL Data Warehouse**. This architecture ensured consistency across the analytical workflow while maintaining clear separation between operational data preparation and reporting activities.

3.2 Data Extraction and Initial Loading

The DataCo Supply Chain dataset was provided in CSV format and imported into Python using the Pandas library. The initial loading phase involved importing the dataset into a Pandas DataFrame, inspecting dataset dimensions, verifying column names and data types, identifying missing values and anomalies, and performing exploratory analysis to understand attribute distributions.

The dataset contained exactly **180,516 records** with more than fifty original attributes encompassing customer, product, logistics, and financial information. The initial inspection confirmed that the dataset was suitable for large-scale analytical processing while highlighting several preprocessing challenges that required resolution prior to database loading.

3.3 Data Cleaning and Preprocessing

Data cleaning represented a critical step and included format adjustments, column alignment, and the replacement of spaces with underscores. This naming convention improved maintainability across Python, SQL, and Power BI environments.

Removal of Redundant Attributes: Several fields provided duplicate information or contained limited analytical value. Redundant columns were removed to improve storage efficiency and simplify dimensional modeling.

Numerical Data Validation: Financial and quantity-related variables (sales, product prices, discounts, profits, and quantities) were validated to ensure numerical consistency. This validation prevented aggregation errors during dashboard development.

3.4 Handling Missing Values and Data Quality Issues

Missing values can significantly affect analytical accuracy and therefore required systematic evaluation. Each variable was individually assessed based on the percentage of missing values, business importance, and feasibility of imputation.

Scenario	Treatment Strategy
Small number of missing values	Removal
Critical numerical values	Imputation
Non-essential categorical fields	Retention as "Unknown"

Duplicate Validation: Duplicate records can distort aggregate metrics and financial calculations. Duplicate checks were performed to ensure uniqueness of order records, consistency of customer identifiers, and integrity of product mappings. The validation process confirmed that the transactional dataset maintained acceptable levels of uniqueness for analytical purposes.

Character Encoding Issues: One significant challenge encountered during preprocessing involved character encoding incompatibilities. The dataset contained non-ASCII characters originating primarily from European product names, accented characters, and special symbols. These encoding issues caused import failures during MySQL loading. The problem was resolved by converting the dataset into an ASCII-compatible format prior to database import. This step ensured compatibility across all downstream architectures.

3.5 Date Transformation and Temporal Features

The original dataset stored date information as text strings. To enable time-series analysis, date attributes were converted into proper datetime formats. The transformed fields supported the extraction of several temporal dimensions:

- `order_year`
- `order_month`
- `order_month_name`
- `order_quarter`
- `order_day_of_week`

These features significantly improved the flexibility of trend analysis and dashboard filtering.

3.6 Feature Engineering

Feature engineering represented the most significant contribution of the data engineering stage. Rather than relying solely on raw operational variables, additional business-oriented attributes were generated to support decision-making.

Formula: $shipping_delay_days = days_for_shipping_real - days_for_shipment_scheduled$

Purpose & Value: Measures deviation between actual and planned delivery performance to support logistics monitoring and delivery risk assessment.

Delivery Performance Classification: The shipping delay metric was transformed into business-friendly delivery categories: *Early Delivery*, *On-Time Delivery*, and *Delayed Delivery* to improve executive interpretation of logistics performance.

- **Delayed Flag:** A binary feature was introduced (1 = delayed, 0 = not delayed) to simplify KPI calculations and dashboard aggregation.
- **On-Time Flag:** A complementary indicator identifying orders delivered according to schedule, supporting service-level monitoring.
- **Severe Delay Flag:** Orders exceeding predefined delay thresholds were classified as severe delays, allowing management to focus on critical operational failures rather than minor deviations.
- **High Discount Flag:** Orders exceeding predefined discount thresholds were classified as high-discount transactions to support pricing strategy evaluation and margin analysis.
- **Loss Order Flag:** Orders generating negative profit were classified as loss-making orders to enable identification of commercially unsustainable transactions.
- **Profit Category:** Profit values were categorized into *High Profit*, *Medium Profit*, *Low Profit*, and *Loss Making* to improve profitability segmentation and portfolio analysis.
- **Shipment Risk Level:** Delivery risk was transformed into categorical risk bands: *Low Risk*, *Medium Risk*, and *High Risk* to support proactive logistics management.
- **Sales Band:** Orders were segmented according to transaction value into *Low Value Orders*, *Medium Value Orders*, and *High Value Orders* to support customer segmentation and revenue concentration analysis.
- **High Value Order Flag:** Large transactions were identified using predefined thresholds to support customer prioritization and premium service allocation.

3.7 Data Export and Database Preparation

Following completion of preprocessing and feature engineering, the analytical dataset was exported in CSV format for loading into MySQL. Several export-related challenges were encountered: character encoding

issues, large file size, and import performance limitations. These challenges were resolved through format standardization and optimized export procedures. The final engineered dataset served as the foundation for the dimensional warehouse developed in the following chapter.

3.8 Chapter Summary

This chapter described the transformation of raw transactional data into an analytical dataset suitable for business intelligence applications. Data cleaning, quality validation, temporal transformations, and feature engineering significantly enhanced the analytical value of the original dataset. The resulting dataset formed the basis for the dimensional warehouse and dashboard architecture developed in subsequent chapters.

CHAPTER 4: DATA WAREHOUSE DESIGN AND DIMENSIONAL MODELING

4.1 Introduction to Data Warehousing

A data warehouse is a centralized repository designed specifically for analytical workloads and decision support systems. Unlike operational databases, which are optimized for transactional processing, data warehouses are optimized for aggregation, reporting, and historical analysis. Operational systems such as Enterprise Resource Planning (ERP), Warehouse Management Systems (WMS), and Customer Relationship Management (CRM) platforms are designed primarily for recording business transactions. These systems prioritize data consistency and transactional integrity rather than analytical performance.

As organizations accumulate increasing amounts of operational data, direct analysis on transactional systems becomes inefficient due to complex table structures, high query latency, excessive joins, and inconsistent business definitions. Data warehouses solve these problems by organizing information logically for strategic queries.

4.2 Importance of Integrated Data Analysis

Analyzing functional business domains independently often produces fragmented insights and limits management visibility. For example, the sales team may focus on revenue growth while logistics teams focus on delivery performance and finance teams focus on profitability. Without integration, it becomes difficult to understand cross-functional relationships, such as:

- Are delivery delays affecting customer retention?
- Do discounts improve sales sufficiently to justify margin erosion?
- Which markets generate revenue but experience operational inefficiencies?
- Which product categories create demand without generating profitability?

A centralized warehouse enables these questions to be answered through integrated analysis.

4.3 Dimensional Modeling

Dimensional modeling is one of the most widely adopted approaches for analytical database design. Unlike highly normalized transactional databases, dimensional models organize information into clear **Fact tables** and **Dimension tables**.

Characteristics of fact tables include a large number of records, numerical measures, foreign keys referencing dimensions, and fine transactional granularity. The central fact table developed in this project is `fact_orders`, which represents customer order transactions.

Grain of the Fact Table: The selected grain for this project is **One row per order line item transaction**. This granularity enables flexible customer, product, profitability, logistics, and inventory analysis. Choosing a more aggregated grain would have limited analytical flexibility.

Measures Stored in `fact_orders`:

- *Sales Metrics:* `sales`, `sales_per_customer`, `order_item_total`
- *Quantity Metrics:* `order_item_quantity`
- *Profitability Metrics:* `order_profit_per_order`, `benefit_per_order`, `order_item_profit_ratio`
- *Discount Metrics:* `order_item_discount`, `order_item_discount_rate`
- *Logistics Metrics:* `shipping_delay_days`, `days_for_shipping_real`, `days_for_shipment_scheduled`

4.5 Dimension Tables

Dimension tables provide descriptive context to business transactions. The warehouse consists of four primary dimensions radiating from the center:

1. **Customer Dimension (`dim_customer`):** Primary Key is `customer_id`. Attributes include `customer_segment`, `customer_city`, `customer_state`, `customer_country`, and `customer_zipcode`. Supports customer segmentation and market analysis.
2. **Product Dimension (`dim_product`):** Primary Key is `product_card_id`. Attributes include `product_name`, `category_name`, `department_name`, and `product_price`. Supports product portfolio and category studies.
3. **Geography Dimension (`dim_geography`):** Attributes include `market`, `order_region`, `order_country`, `order_city`, `latitude`, and `longitude`. Supports geographic mapping and regional risk assessment.
4. **Date Dimension (`dim_date`):** Contains time elements such as `order_year`, `order_month`, `order_month_name`, `order_quarter`, and `order_day_of_week` to simplify trend analysis and seasonality comparisons.

4.10 Star Schema Architecture

The warehouse follows a classic star schema architecture. The center of the schema is occupied by `fact_orders`, while the dimension tables (`dim_customer`, `dim_product`, `dim_geography`, and `dim_date`) radiate outward. This structure resembles a star and is visually optimized for analytical clarity.

4.11 Why Star Schema Was Selected

Several alternative designs were explicitly evaluated before finalizing the star schema:

- **Flat Table Design:** Offered simplicity but suffered from heavy data duplication, poor scalability, large storage requirements, and slower analytical query execution over large sets.
- **Snowflake Schema:** Reduced data redundancy through normalization, but introduced additional relational joins that impaired query performance and significantly increased schema complexity.
- **Star Schema:** Provided excellent analytical performance, intuitive design, simplified SQL queries, and native compatibility with Power BI. For these reasons, it was selected as the optimal architecture.

4.12 ETL Workflow

The operational ETL pipeline implemented during the project is summarized as follows: **CSV Dataset** → **Python Data Cleaning** → **Feature Engineering** → **ASCII Conversion** → **MySQL Import** → **Dimension Table Creation** → **Fact Table Construction** → **Business KPI Queries** → **Power BI Dashboard**. This pipeline mirrors the architecture commonly encountered in standard enterprise analytics environments.

4.13 Data Validation and Reconciliation

Several validation procedures were performed during warehouse construction to guarantee absolute numbers matched across environments:

- Raw Dataset Records: 180,516
- Fact Table Records (`fact_orders`): 180,516
- Customer Dimension Records (`dim_customer`): 20,649
- Product Dimension Records (`dim_product`): 118
- Geography Dimension Records (`dim_geography`): 259
- Date Dimension Records (`dim_date`): 648

4.14 Chapter Summary

This chapter described the design and implementation of the analytical data warehouse used throughout the project. The adoption of dimensional modeling and star schema architecture improved analytical flexibility, simplified dashboard development, and enabled efficient execution of business queries. The resulting warehouse serves as the foundation for the SQL analytical layer.

CHAPTER 5: SQL ANALYTICS LAYER AND BUSINESS INTELLIGENCE QUERIES

5.1 Role of SQL in the Analytics Pipeline

Within this architecture, SQL served several important functions: validation of transformed datasets, KPI generation, business query execution, aggregation of transactional data, and the preparation of clean data exports for visualization. Rather than treating SQL as a passive storage layer, it was used as an active component of the analytical workflow.

5.2 Importance of SQL in Supply Chain Analytics

Supply chain decision-making requires frequent aggregation across multiple dimensions including time, geography, products, customers, and logistics operations. SQL is particularly well suited for these operations due to its ability to efficiently perform grouping, aggregation, filtering, ranking, and joining. Key business questions addressed through SQL included tracking which markets generate the highest revenue, identifying shipping modes with the largest delays, and calculating the exact percentage of orders delivered on time to build dashboard metrics.

5.3 KPI Development Strategy

The project adopted a layered KPI design methodology spanning multiple operational and commercial areas:

- **Operational KPIs:** Measure execution efficiency (e.g., Average Shipping Delay, On-Time Delivery Percentage, Late Delivery Risk, Severe Delay Percentage).
- **Commercial KPIs:** Measure revenue generation (e.g., Total Sales, Average Order Value, Revenue Per Customer).
- **Customer KPIs:** Measure customer behavior (e.g., Total Customers, Customer Segment Contribution, Market Distribution).
- **Product KPIs:** Measure demand characteristics (e.g., Total Units Sold, Average Product Price, Demand Concentration).
- **Financial KPIs:** Measure profitability (e.g., Total Profit, Profit Margin, Loss-Making Orders, Discount Rates).

5.4 Revenue Analytics Queries

One of the first analytical requirements involved measuring organizational revenue performance.

```
-- Query to calculate Total Sales
SELECT
    SUM(sales) AS total_sales
FROM fact_orders;
```

Business Interpretation: This metric represents total organizational revenue generated during the analysis period and forms the foundation for executive performance reporting.

```
-- Query to calculate Revenue by Market
SELECT
    market,
    SUM(sales) AS total_sales
FROM fact_orders
GROUP BY market
ORDER BY total_sales DESC;
```

Business Interpretation: This query identifies geographical markets contributing the highest revenue and supports regional investment decisions.

5.5 Customer Analytics Queries

Customer analytics focused on segmentation and contribution analysis.

```
-- Query to count Total Distinct Customers
SELECT
    COUNT(DISTINCT customer_id)
FROM dim_customer;
```

```
-- Query to calculate Revenue by Customer Segment
SELECT
    customer_segment,
    SUM(sales) AS total_sales
FROM fact_orders f
JOIN dim_customer c ON f.customer_id = c.customer_id
GROUP BY customer_segment;
```

Business Interpretation: This query identifies customer groups contributing disproportionately to revenue generation.

5.6 Logistics Analytics Queries

Logistics performance represented one of the core objectives of the control tower.

```
-- Query for Average Shipping Delay
SELECT
    AVG(shipping_delay_days)
FROM fact_orders;
```

```
-- Query for Shipping Mode Analysis
SELECT
    shipping_mode,
    AVG(shipping_delay_days) AS avg_delay
FROM fact_orders
GROUP BY shipping_mode;
```

```
-- Query for Late Delivery Risk Percentage
SELECT
    ROUND(
        SUM(late_delivery_risk) * 100.0 / COUNT(*),
        2
    ) AS risk_percentage
FROM fact_orders;
```

Business Interpretation: This KPI quantifies the proportion of shipments exposed to delivery risk.

5.7 Product Analytics Queries

Demand analysis focused on product movement and category concentration.

```
-- Query for Top Products by Volume
SELECT
    product_name,
    SUM(order_item_quantity) AS total_units
FROM fact_orders
GROUP BY product_name
ORDER BY total_units DESC
LIMIT 10;
```

```
-- Query for Category Demand Analysis
SELECT
    category_name,
    SUM(order_item_quantity)
FROM fact_orders
GROUP BY category_name;
```


Business Interpretation: These queries support inventory allocation decisions and replenishment planning.

5.8 Financial Analytics Queries

Profitability analysis represented one of the most important business objectives.

```
-- Query for Total Profit
SELECT
    SUM(order_profit_per_order)
FROM fact_orders;
```

```
-- Query for Profit Margin
SELECT
    ROUND(
        SUM(order_profit_per_order) * 100 / SUM(sales),
        2
    ) AS profit_margin
FROM fact_orders;
```

```
-- Query for Loss-Making Orders
SELECT COUNT(*)
FROM fact_orders
WHERE loss_order_flag = 1;
```

Business Interpretation: These metrics identify products, markets, and transactions that generate revenue while destroying financial value.

5.9 Validation Queries

Validation represented a critical component of the analytical workflow. Several reconciliation checks were performed during development to ensure counts were completely identical (e.g., `SELECT COUNT(*) FROM fact_orders;` yielding 180,516, `dim_customer` returning 20,649, `dim_product` returning 118, `dim_geography` returning 259, and `dim_date` returning 648).

5.10 Why SQL Instead of Power BI Alone?

Several calculations could have been performed directly in Power BI using DAX, however, SQL was preferred for several reasons:

- **Scalability:** Database engines are natively optimized for large scale aggregation operations.
- **Reusability:** SQL logic can be reused seamlessly across multiple different reporting or data science tools.

- **Transparency:** Business users can audit standard SQL queries more easily than complex, nested DAX expressions.
- **Performance:** Pre-aggregated warehouse queries significantly reduce front-end dashboard refresh times.
- **Industry Practice:** Enterprise architectures commonly perform structural transformations within core databases rather than visual layers.

5.11 Export Strategy

After validation, warehouse tables were exported into CSV format and transferred to the Power BI environment. The exported tables included `fact_orders`, `dim_customer`, `dim_product`, `dim_geography`, and `dim_date`. This approach provided several advantages: reduced dependency on active database connectivity, simplified dashboard portability, and faster dashboard refresh times.

5.12 Chapter Summary

This chapter demonstrated how SQL was used as an analytical engine supporting the Supply Chain Control Tower. Business KPIs were implemented through SQL queries and validated through reconciliation checks before dashboard development.

CHAPTER 6: DASHBOARD DESIGN AND IMPLEMENTATION

This chapter describes the design philosophy, KPI selection methodology, visualization choices, and business questions addressed by the Supply Chain Control Tower dashboard. The objective of the dashboard was not merely to display data but to create an integrated decision-support system capable of providing visibility across multiple supply chain functions. The dashboard was designed according to three core principles: Executive readability, Operational visibility, and Actionable insights. The final solution consisted of five analytical pages, each addressing a distinct business domain.

6.1 Dashboard Design Philosophy

Traditional reports often require users to navigate multiple systems and manually consolidate information before making decisions. The purpose of the control tower was therefore to provide centralized visibility adapted to the needs of different stakeholders (Logistics teams look at operations, sales teams look at customers, category managers look at products, and finance leadership looks at margins).

6.2 Dashboard Architecture

The dashboard architecture follows a strict hierarchical analytical structure: **Executive Performance** → **Operational Performance** → **Customer Insights** → **Product Demand** → **Financial Outcomes**. This

progression mirrors the decision-making process within most organizations, allowing executives to begin with high-level performance metrics before drilling down into operational drivers and profitability factors.

6.3 Page 1 — Executive Overview

The Executive Overview page serves as the entry point of the control tower and provides a high-level summary of organizational performance, answering total revenue, profitability, and volume questions over time.

- **Key Performance Indicators:** Total Sales (Revenue measurement), Total Profit (Profitability measurement), Total Orders (Transaction volume), Total Customers (Customer base size), and Profit Margin (Financial efficiency).
- **Visualizations Used:** Annual Sales Trend (identifies growth trends and seasonality), Annual Profit Trend (evaluates financial sustainability), Revenue by Market (identifies geographical growth opportunities and market concentration risk), and Revenue by Customer Segment (supports commercial strategy).

Key Findings: The dashboard revealed significant geographical differences in revenue generation and customer contribution. Several markets demonstrated disproportionately high revenue contributions, highlighting clear strategic priorities.

6.4 Page 2 — Logistics and Delivery Performance

Logistics performance directly influences customer satisfaction and operational costs. This dashboard page focuses on delivery efficiency, transit duration, and transportation risk.

- **Key Performance Indicators:** Average Shipping Delay, On-Time Delivery Percentage, Late Delivery Risk, Severe Delay Percentage, and Average Shipping Days.
- **Visualizations Used:** Delivery Performance Distribution (Classifies orders as early, on-time, or delayed), Shipping Mode Analysis (Compares delivery performance across Same Day, First Class, Second Class, and Standard Class), Market Risk Analysis (Identifies regions experiencing elevated operational risk), and Monthly Delay Trend (Supports seasonality analysis and logistics planning).

Key Findings: The analysis revealed that while a large proportion of shipments carried delivery risk, the average delay duration remained relatively small. This suggests that operational disruptions were frequent but generally manageable.

6.5 Page 3 — Customer and Market Analytics

Customer analytics focuses on understanding demand sources, market reach, and revenue contributors across segments.

- **Key Performance Indicators:** Total Customers, Revenue per Customer, Average Order Value, Corporate Revenue Share, and Consumer Revenue Share.

- **Visualizations Used:** Revenue by Customer Segment (Supports pricing and marketing decisions), Customer Count by Segment (Identifies concentration risk), Geographic Revenue Distribution (Visibility into international market performance), and Top Countries by Revenue (Highlights high-value markets).

Key Findings: Customer contribution was not evenly distributed across segments or regions, emphasizing the high importance of segmentation-driven strategies.

6.6 Page 4 — Product and Inventory Analytics

The Product and Inventory dashboard investigates product demand, velocity, and movement patterns.

- **Key Performance Indicators:** Total Products, Average Units per Order, High Demand Products Percentage, Average Product Price, and Total Units Sold.
- **Visualizations Used:** Demand by Category (Identifies categories requiring inventory prioritization), Revenue vs Volume Analysis (Evaluates tradeoffs), Top Products by Units Sold (High-velocity inventory items), and Sales Band Distribution (Low-value, medium-value, and high-value transactions).

Key Findings: Demand concentration analysis revealed that a relatively small subset of products generated a disproportionately large share of movement, supporting targeted inventory management strategies.

6.7 Page 5 — Profitability and Financial Analytics

Revenue growth alone does not guarantee financial success. This dashboard page focuses on uncovering margin leakages and tracking key financial metrics.

- **Key Performance Indicators:** Total Profit, Profit Margin, Loss-Making Orders, Average Discount Rate, and Profit per Order.
- **Visualizations Used:** Profit by Category (Supports portfolio optimization), Discount versus Profit Analysis (Evaluates the relationship between pricing incentives and profitability), Profit Category Distribution (Classifies transactions by contribution), Top Products by Profit (High-value assets), and Loss-Making Orders by Category (Highlights areas requiring immediate corrective action).

Key Findings: The analysis demonstrated that high sales volumes do not necessarily translate into profitability. Several products and categories generated substantial revenue while contributing limited or negative financial value due to aggressive discounting.

6.8 Dashboard Interactivity

The dashboard includes extensive interactive capabilities including year filters, market filters, customer segment filters, drill-down analysis, and cross-visual filtering. These features enable users to transition seamlessly from executive summaries to detailed operational insights.

6.9 Benefits of the Dashboard Approach

Compared with static reports, the control tower provides faster decision-making, integrated cross-functional visibility, proactive issue identification, reduced manual reporting effort, and improved stakeholder communication.

6.10 Chapter Summary

This chapter described the design and implementation of the Supply Chain Control Tower dashboard. The dashboard transformed warehouse data into actionable insights across executive, operational, customer, product, and financial dimensions. By integrating multiple business perspectives into a unified analytical platform, the control tower supports both strategic planning and operational decision-making.

CHAPTER 7: BUSINESS INSIGHTS AND RECOMMENDATIONS

7.1 Executive Summary of Findings

The analysis identified several critical operational themes across logistics, customer behavior, product demand, and profitability. The major findings include: revenue generation was concentrated in a limited number of markets; delivery risks were widespread despite relatively small average delays; demand was heavily concentrated among a relatively small subset of products; high sales volumes did not always translate into profitability; and aggressive discounting strategies occasionally resulted in clear value destruction. These findings demonstrate the importance of integrated decision-making across supply chain functions.

7.2 Logistics and Delivery Insights

Finding 1: Delivery Risk Is Substantially Higher Than Expected. The analysis revealed that a significant proportion of orders were associated with late delivery risk indicators. Although the average delay duration remained relatively small, the high frequency of delivery risk events can lead to reduced brand loyalty over time. Even minor delays become significant when experienced repeatedly across a large customer base.

Recommendation: The organization should establish proactive logistics monitoring systems capable of identifying at-risk shipments before service failures occur. Potential initiatives include predictive delivery risk models, dynamic route optimization, carrier performance monitoring, and automated exception alerts.

7.3 Shipping Mode Performance Insights

Finding 2: Premium Shipping Options Do Not Always Deliver Superior Performance. Analysis of shipping modes indicated that premium shipping services did not consistently outperform standard delivery options. In certain cases, lower-cost shipping methods demonstrated more stable delivery performance than

premium alternatives. Organizations frequently assume premium services provide superior reliability, but inefficient route planning or poor carrier utilization can reverse this benefit.

Recommendation: The organization should regularly evaluate shipping cost, transit reliability, customer satisfaction, and delivery consistency. Shipping mode selection should be optimized according to total service value rather than delivery speed alone.

7.4 Customer and Market Insights

Finding 3: Revenue Concentration Exists Across Markets and Customer Segments. Revenue contribution analysis demonstrated that a relatively small number of markets generated a disproportionately large share of organizational revenue. Similarly, customer segments contributed unevenly to overall business performance. Revenue concentration increases exposure to regional economic disruptions and market-specific risks.

Recommendation: Management should pursue diversification strategies by expanding underrepresented markets, improving penetration within emerging regions, and developing targeted commercial accounts.

7.5 Product Demand Insights

Finding 4: Demand Is Highly Concentrated Among Selected Products. The product demand analysis revealed that a limited number of products generated a significant proportion of overall sales volume. This pattern is consistent with the Pareto Principle commonly observed in retail networks. Demand concentration creates operational challenges including stockouts, warehouse congestion, replenishment pressure, and supplier dependency.

Recommendation: Organizations should classify products according to demand characteristics using approaches such as ABC analysis, inventory segmentation, and demand forecasting models. Priority inventory allocation should be assigned to high-demand products.

7.6 Inventory Management Insights

Finding 5: Inventory Optimization Opportunities Exist Despite Missing Stock Data. Although direct inventory information was unavailable within the dataset, demand patterns provided valuable proxies for inventory planning. Products exhibiting consistently high movement volumes represent critical inventory categories requiring priority management attention. Without inventory prioritization, organizations may experience excessive holding costs or stock shortages.

Recommendation: Future implementations should integrate warehouse stock levels, safety stock calculations, reorder points, and inventory turnover metrics. The current control tower provides a strong foundation for future inventory optimization initiatives.

7.7 Financial and Profitability Insights

Finding 6: Revenue Growth Does Not Guarantee Profitability. The profitability analysis demonstrated that several products and categories generated substantial revenue while contributing limited financial returns. Revenue-focused decision-making can create misleading perceptions of business health. Products generating large sales volumes may consume operational resources without producing adequate returns.

Recommendation: Organizations should supplement revenue metrics with contribution margin analysis, product profitability analysis, and customer lifetime value calculations. Management decisions must prioritize profitability rather than sales volume alone.

7.8 Discount Strategy Insights

Finding 7: High Discounts Can Destroy Value. The relationship between discount rates and profitability demonstrated diminishing returns beyond certain discount thresholds. While moderate discounts stimulated demand, excessive discounting frequently reduced profitability disproportionately, eroding margins and training customers to wait for promotions.

Recommendation: Organizations should adopt data-driven pricing strategies based on customer elasticity, competitive positioning, inventory objectives, and profitability targets. Discount approval processes should include profitability impact assessments.

7.9 Strategic Recommendations Summary

Focus Area	Core Strategic Recommendation
Logistics	Implement proactive, predictive delivery monitoring systems
Inventory	Introduce structural demand forecasting models
Markets	Pursue diversification to reduce geographical concentration risk
Products	Prioritize high-demand SKUs across fulfillment lanes
Finance	Monitor core profitability metrics alongside gross revenue
Pricing	Introduce unified discount governance frameworks

7.10 Managerial Implications

Historically, organizations have managed logistics, sales, customer management, inventory, and finance as separate operational silos. The Supply Chain Control Tower demonstrates that these functions are deeply interconnected (e.g., logistics performance influencing customer retention, discounts affecting profitability, and demand patterns influencing inventory requirements). An integrated analytical approach enables organizations to optimize overall system performance rather than individual departmental objectives.

7.11 Chapter Summary

This chapter transformed analytical observations into managerial recommendations capable of improving operational and financial performance. The findings demonstrate that data-driven decision-making can significantly improve visibility, responsiveness, and resilience within modern supply chains.

CHAPTER 8: FUTURE SCOPE AND CONCLUSION

8.1 Future Scope

The implemented control tower primarily focuses on descriptive and diagnostic analytics. Future versions can evolve toward predictive and prescriptive analytics, enabling organizations not only to understand what happened but also to anticipate future events and recommend optimal actions.

8.1.1 Demand Forecasting

The current dashboard provides historical demand visibility through sales and order analysis. However, supply chain organizations require forward-looking forecasts to support procurement, inventory allocation, and production planning. Machine learning algorithms could be introduced to forecast daily, weekly, and monthly demand fluctuations using Moving Average Models, Exponential Smoothing, ARIMA, Prophet, Gradient Boosting, or Long Short-Term Memory Networks (LSTM).

8.1.2 Inventory Optimization

Integrating inventory data would allow the calculation of optimal safety stock levels, carrying costs, and reorder points. Potential metrics include Days of Inventory Outstanding, Inventory Turnover Ratio, Fill Rate, and Stockout Frequency, transforming the dashboard into a comprehensive inventory control platform.

8.1.3 Predictive Delivery Risk Models

Future versions could use historical data and features (shipping mode, destination market, order size, carrier performance) to build predictive classification algorithms (Random Forests, XGBoost) to estimate delay probabilities before shipments leave the warehouse, enabling proactive interventions.

8.1.4 Supplier Performance Analytics

Expanding the analytical data model to include procurement costs and supplier reliability metrics would significantly improve purchasing decisions. Key metrics include Supplier On-Time Delivery Rate, Supplier Fill Rate, and Supplier Defect Rate.

8.1.5 Route Optimization

Incorporating route optimization algorithms, vehicle utilization analysis, fuel cost tracking, and carrier selection models via Linear Programming or Genetic Algorithms could dramatically reduce transportation expenditure while maintaining service quality.

8.1.6 Real-Time Data Integration

The current control tower operates using batch processing and historical data. Future implementations could incorporate streaming architectures (e.g., Apache Kafka, Apache Flink) capable of processing live operational data to support instant alerts, dynamic risk monitoring, and live tracking.

8.1.7 Artificial Intelligence and Prescriptive Analytics

The next logical evolution involves prescriptive decision-making, such as automatic inventory allocation recommendations, automated shipment rerouting suggestions, pricing optimization engines, and active customer prioritization models.

8.2 Project Contributions

This project contributes in several important ways. First, it demonstrates the integration of multiple analytical technologies into a single end-to-end solution combining Python, MySQL, and Power BI. Second, it establishes a robust feature engineering framework that populates metrics such as shipping delay fields and profitability flags. Third, it demonstrates the practical application of dimensional modeling and star schema design within supply chain analytics environments.

8.3 Achievement of Project Objectives

Objective Description	Status
Develop analytical pipeline	Completed
Perform feature engineering	Completed
Build dimensional warehouse	Completed
Develop executive dashboards	Completed
Generate business insights	Completed
Provide recommendations	Completed

8.4 Limitations of the Study

Despite its strengths, several limitations remain. The dataset lacked direct inventory tracking levels, supplier info, procurement data, and transportation cost fields, restricting the depth of certain optimization layers. Furthermore, the data was historical in nature, meaning real-time streaming was out of scope. Finally, the available data for 2018 extended only until January, preventing meaningful annual year-over-year trend calculations involving that specific cycle.

8.5 Conclusion

The increasing complexity of modern supply chains has created an urgent need for integrated analytical solutions capable of improving visibility, responsiveness, and decision quality. This project addressed these challenges through the development of an end-to-end Supply Chain Control Tower integrating data engineering, warehousing, business intelligence, and analytical reporting.

The resulting platform successfully transformed raw transactional data into actionable business insights across five critical domains: executive performance, logistics operations, customer behavior, product demand, and financial outcomes. Effective supply chain management is no longer driven by intuition alone but increasingly by integrated, data-driven decision-making. As organizations continue to face rising complexity and uncertainty, analytical control towers will become an increasingly important component of competitive advantage.

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APPENDICES

Appendix A: SQL Scripts

- Schema creation scripts
- Dimension table scripts
- Fact table scripts
- KPI queries

Appendix B: Python Feature Engineering Scripts

- Data cleaning scripts

- Feature engineering code
- Export procedures

Appendix C: Power BI DAX Measures

- Revenue calculations
- Profit calculations
- Margin calculations
- Customer KPIs

Appendix D: Dashboard Screenshots

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- Page 2: Logistics and Delivery
- Page 3: Customer Analytics
- Page 4: Product Analytics
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